

Arshia Fazeli

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Education

University of Waterloo

Sept 2021 – Sept 2026

PhD in Chemical Engineering

- GPA : 92.67/100
- **Coursework:** Theory and Application of Transport Phenomena, Applied Engineering Mathematics, Chemical Reactor Analysis, Computational Fluid Dynamics for Engineering Design, Multiphase Flows, Computational Fluid Dynamics, Consulting

University College London

Sept 2018 – June 2021

BEng in Chemical Engineering

- First Class Honour
- **Coursework:** Computational Modelling and Analysis, Reaction Engineering, Process Dynamics and Control, Particulate Systems and Separation Processes, Programming, Manufacture of Regenerative Medicine Products

Experience

Computational Multiphysics Researcher

Sept 2021 – Now

University of Waterloo

- Formulated and validated a well-posed Two-Fluid Model (TFM) incorporating a laminar dispersion force closure to broaden the hyperbolicity domain for dispersed multiphase flows.
- Implemented discontinuous Galerkin (DG) finite element solvers in Python/NGSolve for coupled multiphysics systems: two-fluid models, incompressible Navier–Stokes, and Cahn–Hilliard Navier–Stokes two-phase flow.
- Verified solver accuracy via the method of manufactured solutions, quantifying convergence rates and discretization error.

Teaching Assistant

Sept 2021 – Now

University of Waterloo

- Led tutorials and office hours for graduate courses in transport phenomena, numerical methods for ODEs/PDEs, and computational methods.
- Engineered a randomized Python unittest autograder for a programming course — test inputs are freshly generated each run to prevent hardcoded solutions, with a weighted scoring system, safe student code execution via `runpy`, and per-student log files with filtered, line-level error feedback.

Research Associate

Aug 2020 – Dec 2020

University College London

- Conducted molecular dynamics simulations using LAMMPS and PLUMED to investigate nucleation pathways in colloidal suspensions.

Projects

Discontinuous Galerkin based solvers for multiphysics problems

- Implemented DG finite element solvers for a range of multiphysics problems: Cahn–Hilliard equation, incompressible Navier–Stokes, mixture model, Euler–Euler two-fluid model, and monolithic Cahn–Hilliard Navier–Stokes (CHNS) for two-phase flows.
- Implemented adaptive mesh refinement driven by a Zienkiewicz-Zhu (ZZ) gradient-recovery error estimator, enabling targeted refinement in regions of large solution gradients.
- Verified numerical accuracy via the method of manufactured solutions, confirming optimal convergence rates.
- Tools Used: Python, NGSolve, NETGEN, PETSc, GMSH, ParaView

LLM-Generated Code Detector

- Built an end-to-end ML pipeline for detecting AI-generated student code; engineered 18 AST-based and stylistic features (Halstead metrics, Shannon entropy, comment semantics, call nesting depth) and trained a Random Forest classifier via 5-fold stratified cross-validation with `RandomizedSearchCV`, achieving ROC-AUC of 0.95.
- Implemented a TF-IDF cosine similarity plagiarism checker to flag near-identical submission pairs across a student cohort.
- Tools Used: Python, scikit-learn, pandas, NumPy

Steady Incompressible Navier-Stokes Solvers with Scalar Transport and Reaction [SCGS](#) [SIMPLE](#)

- Implemented finite-volume solvers for the incompressible Navier-Stokes equations with SIMPLE and SCGS velocity-pressure coupling; leveraged Numba JIT compilation for significant runtime speedups.
- Extended the SIMPLE solver to include scalar transport and reaction source terms.
- Implemented multiple convection and diffusion discretization schemes.
- Tools Used: Python, NumPy, Numba, Matplotlib, SymPy

Euler Equations Solver Using Flux Vector Splitting [EULER-FVS](#)

- Implemented a finite-volume method solver for the one-dimensional compressible Euler equations using multiple flux splitting schemes: Steger-Warming, Van Leer, and AUSM+ (Liou 1996), the latter employing an advection/pressure operator-split formulation for improved accuracy on contact discontinuities and shocks.
- Tools Used: Python, NumPy, SymPy, Matplotlib

CFD Simulation of a Side-Dump Combustor with Variable Inlet Angles

- Performed CFD simulations of a turbulent airflow in a side-dump combustor.
- Validated simulation results against experimental velocity and turbulence data.
- Investigated the impact of side-inlet angle on fuel-air mixing.
- Conducted mesh refinement and error analysis to quantify discretization and modelling errors.
- Tools Used: Ansys CFX

Publication

Laminar dispersion force effects on two-fluid modelling and simulation of bubble column hydrodynamics [link](#)
Arshia Fazeli, Sander Rhebergen, Nasser Mohieddin Abukhdeir

Volunteering

VP Academic at University of Waterloo Chemical Engineering Graduate Student Association (CEGSA)	<i>Jan 2022 - Jan 2023</i>
Welfare Officer at UCL Technology Society	<i>Aug 2020 - July 2021</i>
Outreach Volunteer at Engineers Without Borders	<i>Oct 2019 - Apr 2020</i>
Engineer at UCL Project Impactive	<i>Oct 2019 - Jan 2020</i>

Awards

CSChE 2024 Best Poster Award	certificate , Nov 2024
Garry L. Rempel Memorial Graduate Scholarship	<i>May 2022</i>
Best Final Design Project Team Award	<i>June 2021</i>

Technologies

Languages: Python, MATLAB, C++

Tools: NumPy, Numba, scikit-learn, NGSolve, PETSc, OpenFOAM, Ansys CFX, ParaView, Git, L^AT_EX, Linux, unittest, pytest